

Radar-Based Human Gait Recognition Using Dual-Channel Deep Convolutional Neural Network

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Abstract—This paper addresses the problem of radar-based human gait recognition based on the dual-channel deep convolutional neural network (DC-DCNN). To enrich the limited radar data set of human gaits and provide a benchmark for classifier training, evaluation, and comparison, it proposes an effective method for radar echo generation from the infrared, publicly accessible motion capture (MOCAP) data set. According to the different nonstationary characteristics of micro-Doppler (m-D) for the torso and limbs, it enhances their distinguishable joint time–frequency (JTF) features by applying the short-time Fourier transforms (STFTs) with varying sliding window length and then designs the DC-DCNN structure to achieve refined human gait recognition by separate feature extraction and fusion. Experiments have shown that compared with the traditional single-channel deep convolutional neural network (SC-DCNN), the proposed method achieves higher recognition accuracy in refined human gait classification without incurring additional radar resources and could be readily extended to refined recognition of other human activities.

Index Terms—Deep convolutional neural networks (DCNNs), dual-channel, human gait recognition, motion capture (MOCAP) data set, short-time Fourier transform (STFT).

I. INTRODUCTION

A. Problem Statement

THANKS to the all-time, all-weather, and long-range operational abilities, radar-based human gait recognition has played significant roles in abnormal behavior detection [1], medical assistance [2], [3], military fields [4], and so on. In addition to the bulk motion induced by the torso, some human structures, e.g., the limbs, are rotating or swinging during observation, which could induce additional time-varying frequency modulations on radar echoes, known as the micro-Doppler (m-D) [5], [6]. Because different m-D signatures usually correspond to particular human gaits,

they serve as effective features for radar-based human gait recognition. In recent years, m-D-based human gait recognition has received intensive attention in the radar automatic target recognition community [7]–[18], [28]–[32].

To design feature extractors and classifiers successfully, a large number of training samples, i.e., various radar m-D signatures, are necessary. Unfortunately, publicly accessible data sets are unavailable for various reasons such as individual privacy and network limitations. In addition, the cost of recording a large amount of measured data is relatively high considering the limited experimental environment, hardware conditions, human resources, time consumption, and data processing problems. On the other hand, optical and infrared data sets of various human gaits, e.g., the motion capture (MOCAP) database, are provided by research institutes like Carnegie Mellon University. By converting these data sets to radar echoes of human gaits in various observation scenarios, we could obtain sufficient benchmark data sets for classifier training, evaluation, and comparison. To achieve this goal, it is necessary to study effective techniques for human motion modeling and radar echo generation. There have been some attempts in the literature to utilize MOCAP data or data from visual/MOCAP to augment the available radar data for classification [5], [19], [20]. For instance, the Kinect sensor, which includes an RGB video camera and an infrared sensor, serves as a markerless system for capturing the time-varying coordinate information of human joints, based on which the instantaneous slant range is calculated [19]. However, the Kinect sensor's depth measurement is not as accurate as specially designed MOCAP systems. In addition, the shadowing effect is an important limitation when applying the Kinect to human m-D simulation because it is possible for the motion of part of the body to shield or hide due to the small number of sensors. Some research studies applying the MOCAP database provided by CMU for radar echo generation; on the other hand, they do not elaborate on the construction of the human motion model and the acquisition of the instantaneous slant range [5], [20].

Because the 1-D time signal or the Doppler spectrum is inadequate in describing the local time–frequency characteristics that are vital for effective recognition, we usually perform the joint time–frequency (JTF) transform to the narrowband signal and then adopt the JTF images as training and test samples. Previous methods for radar-based human gait recognition mainly rely on hand-designed feature extractors and classifiers,

Manuscript received October 30, 2018; revised June 6, 2019; accepted July 12, 2019. Date of publication August 6, 2019; date of current version November 25, 2019. This work was supported in part by the National Natural Science Foundation of China under Grant 61522114 and Grant 61631019, in part by the Foundation for the Author of National Excellent Doctoral Dissertation of China under Grant 201448, and in part by the Fund for Foreign Scholars in University Research and Teaching Programs (111 Project) under Grant B18039. (Corresponding author: Xueru Bai.)

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Digital Object Identifier 10.1109/TGRS.2019.2929096

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which require a large amount of expertise and computational complexity [7]–[18]. Recently, the deep convolutional neural network (DCNN) [21]–[27] has been successfully applied to m-D-based human gait recognition [28]–[32]. Although satisfying results could be obtained, such algorithms ignore different motion periods for the torso and micromotion structures and only perform JTF transform with single fixed window length. To improve the recognition accuracy, further research should be done for effective feature extraction and fusion of various body structures.

B. Prior Work

Traditional m-D-based feature extractors mainly focus on extracting: 1) features in the time, frequency, or JTF domains, e.g., period and Doppler bandwidth of the torso and limbs, and so on [7]–[12] and 2) statistical features of the JTF image, e.g., the variance of the upper and lower envelopes, the entropy, moments, and image directionality [13]. Since then, the principal component analysis (PCA) and independent component analysis (ICA) are applied for feature dimension reduction [14]–[16]. The frequently adopted classifiers mainly include the support vector machine (SVM) [7], [8], [12]–[15], the nearest neighbor (NN) classifier [17], the naive Bayes classifier [10], [11], [18], binary trees [10], [11], and so on. As discussed earlier, traditional gait recognition methods require a large amount of professional knowledge and expertise, which may induce high-computational complexity and poor generalization performance.

In recent years, the DCNN, i.e., one of the typical architectures of deep learning, has been successfully applied to various problems, e.g., image recognition and segmentation [21]–[24], speech recognition [25], optical-based human behavior recognition [26], superresolution imaging [27], and so on. This data-driven structure can automatically learn abstract hierarchical features from raw data, thus shedding light on complex feature extraction and recognition problems such as m-D-based human gait recognition [28]–[32]. Initially, the DCNNs are applied to the recognition of human activities, such as walking, running, jumping, climbing, boxing, and so on, and obtained satisfying results [28], [30], [32]. In addition, Park *et al.* [29] and Kim *et al.* [31] adopt the DCNN trained by the ImageNet [33] to classify five types of underwater motion based on transfer learning.

According to the definitions of JTF transforms, e.g., the short-time Fourier transform (STFT), the time and Doppler resolutions are determined by the sliding window length [34]. Generally, a smaller window length generates higher time resolution and lower Doppler resolution, yielding a better description of strong nonstationary motion like the swinging of limbs. On the contrary, a larger window length generates lower time resolution and higher Doppler resolution, which is suitable for the description of the relatively stable torso motion. Experiments have shown that different sliding window lengths yield different recognition accuracies [35]. Generally, the torso has a longer motion period than the limbs. However, most of the available research studies ignore this characteristic and apply STFT images with a single fixed window length for

recognition. For refined human gaits such as walking with guns and walking with swinging arms, the accuracy of such technique reduces to 34% [30], indicating the incompetence of the available DCNN structures in extracting distinguishable refined features of similar gaits.

C. Contribution

To enrich the training and test samples in radar-based human gait recognition, this paper proposes an effective method for radar echo generation based on the infrared MOCAP data under various conditions. Then, to fully exploit the JTF features of various body segments such as the torso and limbs, we propose the dual-channel DCNN (DC-DCNN). The network effectively implements automatic feature extraction of human gaits based on DC-DCNN and achieves effective refined gait recognition by feature fusion. Compared with the available human gait recognition methods, the key developments of DC-DCNN include the following.

- 1) By adding sliding windows with different lengths, we enhance the JTF features of torso and limbs separately, which then serve as the input of DC-DCNN to facilitate refined feature extraction and recognition of various gaits.
- 2) The dual channels merge the hierarchical features of torso and limbs in the concatenation layer, which could effectively increase the recognition accuracy without adding additional radar resources, i.e., only single radar is required for human gait recognition and there is no need for additional radar data.
- 3) For typical walking gaits in the MOCAP data set, e.g., normal walking, high-leg walking, and bending walking, the average recognition accuracy of DC-DCNN reaches 96.24%, which is about 6% higher than traditional single-channel DCNN (SC-DCNN), which adopts samples generated by a single sliding window with a fixed length. In addition, the proposed network structure is ready to be extended to refined recognition of other human activities such as jumping.

The remainder of this paper is organized as follows. Section II studies the generation of radar echoes of human gaits based on the MOCAP data. Section III discusses the choice of sliding window length and then introduces the structure and training process of DC-DCNN in detail. Section IV presents the recognition results with discussions, and Section V concludes this paper.

II. GENERATION OF HUMAN GAIT ECHOES FROM THE MOCAP DATA

The MOCAP database collected by the Graphics Laboratory at the Carnegie Mellon University [36] was generated from infrared cameras, which record some infrared markers on the human body during various activities. Compared with the empirical mathematical parameterization model that is limited in describing various human gaits, i.e., the Boulic model [37], such database is able to describe the motion characteristics of much more human gaits and complex activities, e.g., crawling, running, jumping, and so on. Accordingly, this paper will study

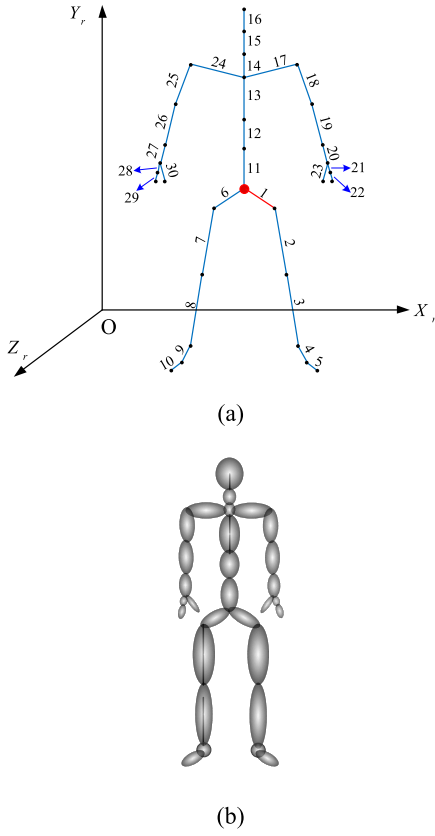


Fig. 1. Human skeleton model for the MOCAP data. (a) Reference points and the indices of bones. (b) Ellipsoid model.

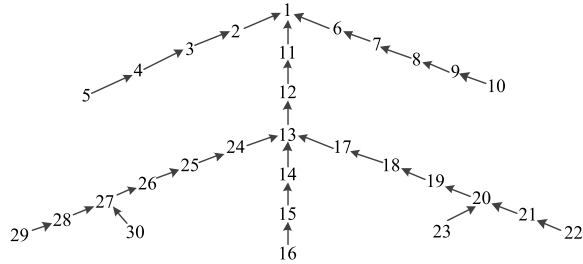


Fig. 2. Dendrogram of bones.

how to generate radar echoes effectively from the MOCAP data set.

A. Human Skeleton and Motion Description Models

As shown by the markers in Fig. 1(a), the human skeleton model adopted by the MOCAP database consists of 31 reference points, where the red dot denotes the root node. A bone is determined by two adjacent reference points, and the model includes 30 bones. For simplicity, each bone is modeled by an ellipsoid, as shown in Fig. 1(b).

Fig. 2 shows the dendrogram of skeleton hierarchy. The index of the root bone is 1, and the i th bone for any $i \in [2, N_b]$ can be traced back to the root bone along the path determined by the pointing direction of arrows, where N_b is the number of bones. The bones that are passed through during the tracing process determine the parent set of the

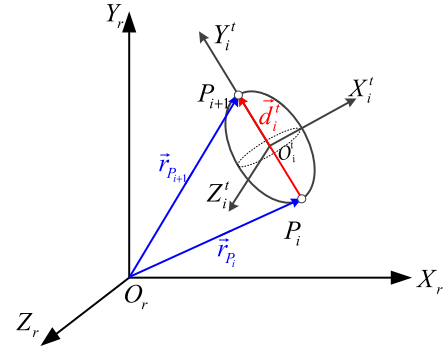


Fig. 3. Local coordinate system of the i th bone at the t th time instant.

i th bone, i.e., $\{n_{i,1}, n_{i,2}, \dots, n_{i,J_i}\}$, where J_i is the number of parent bones for the i th bone. For $i > 1$, we have $n_{i,1} = 1$. For instance, according to Fig. 2, the parent set of the 18th bone is $\{1, 11, 12, 13, 17\}$. For the i th bone, the reference point connected to its adjacent parent bone is defined as the *front endpoint* P_i , while the remaining one is defined as the *post endpoint* P_{i+1} . Accordingly, the front endpoint of the root bone is the root node.

B. Coordinate Systems and Calculation of the Position Vector

The MOCAP data are recorded in the reference coordinate system $X_r Y_r Z_r$ shown in Fig. 1(a), which is fixed and time invariant. To describe the nonrigid motion of the i th bone at time instant t , we establish the local coordinate system $(XYZ)_i^t$, which is a function of t and i . As shown by the right-handed coordinate system in Fig. 3, the Y_i^t -axis is along the direction determined by the center of the ellipsoid O_i^t and the reference point P_i (or P_{i+1}); and the X_i^t - and Z_i^t -axes are along the semimajor axis and semiminor axis of the ellipsoid, respectively.

The direction vector $\vec{d}_i^t$ of the i th bone at t is determined by the two position vectors of the front and post endpoints in the reference coordinate system $X_r Y_r Z_r$, that is

$$\vec{d}_i^t = \vec{r}_{P_{i+1}}^t - \vec{r}_{P_i}^t. \quad (1)$$

Then, the position vector of the i th bone at t , i.e., the position vector of its front endpoint, satisfies

$$(x_{r,i}^t, y_{r,i}^t, z_{r,i}^t)^T = \mathcal{T}_{i,\text{ref}}^t (x_i^t, y_i^t, z_i^t)^T + \vec{m}_i^t \quad (2)$$

where $(\cdot)^T$ is the vector transpose, $(x_i^t, y_i^t, z_i^t)^T$ is the position vector in the $(XYZ)_i^t$ coordinate system, and $\mathcal{T}_{i,\text{ref}}^t$ is the coordinate transform matrix between $(XYZ)_i^t$ and $X_r Y_r Z_r$ at t . In addition, $\vec{m}_i^t$ is the position vector of O_i^t in the $X_r Y_r Z_r$ coordinate system, which is calculated by

$$\vec{m}_i^t = \frac{1}{2} (\vec{r}_{P_{i+1}}^t + \vec{r}_{P_i}^t). \quad (3)$$

In the MOCAP database, we could read $\mathcal{T}_{i,\text{ref}}^t$ and $(x_i^t, y_i^t, z_i^t)^T$ from the files; thus, we focus on the calculation of $\vec{r}_{P_i}^t$ and $\vec{r}_{P_{i+1}}^t$ for $i \in [1, N_b]$, $t \in [1, T]$ in the following discussions.

For $t = 1$, we read the position vector $\vec{r}_{P_1}^1$ of the root bone, i.e., $i = 1$, in the coordinate system $X_r Y_r Z_r$, then the position vector of the post endpoint is calculated as

$$\vec{r}_{P_2}^1 = \vec{r}_{P_1}^1 + \vec{d}_1^1 \quad (4)$$

where $\vec{d}_1^1$ is the direction vector of the root bone and satisfies

$$\vec{d}_1^1 = \mathcal{T}_{1,\text{ref}}^1 [0, L_1, 0]^T. \quad (5)$$

In (5), $\mathcal{T}_{1,\text{ref}}^1$ is the coordinate transform matrix between $(XYZ)_1^1$ and $X_r Y_r Z_r$, and L_1 is the bone length. For other bones, the position vectors $\vec{r}_{P_i}^1$ and $\vec{r}_{P_{i+1}}^1$ can be expressed by

$$\begin{cases} \vec{r}_{P_i}^1 = \vec{r}_{P_1}^1 + \sum_{j=1}^{J_i} \vec{d}_{n_{i,j}}^1 \\ \vec{r}_{P_{i+1}}^1 = \vec{r}_{P_i}^1 + \sum_{j=1}^{J_i} \vec{d}_{n_{i,j}}^1 + \vec{d}_i^1 \end{cases} \quad (6)$$

where $\vec{d}_{n_{i,j}}^1$, $j \in [1, J_i]$, is the direction vector of the $n_{i,j}$ th parent bone in the reference coordinate system.

Then, the direction vector $\vec{d}_i^1$ of the i th bone satisfies

$$\begin{cases} \vec{d}_i^1 = \mathcal{T}_{1,\text{ref}}^1 \prod_{j=2}^{J_i} \mathbf{R}_{n_{i,j}, n_{i,j-1}}^1 [0, L_i, 0]^T \\ \vec{d}_{n_{i,j}}^1 = \mathcal{T}_{1,\text{ref}}^1 \prod_{j'=2}^j \mathbf{R}_{n_{i,j'}, n_{i,j'-1}}^1 [0, L_{n_{i,j}}, 0]^T \end{cases} \quad (7)$$

where $\mathbf{R}_{n_{i,j}, n_{i,j-1}}^1$ is the coordinate transform matrix from the $(XYZ)_{n_{i,j}}^1$ coordinate system of the $n_{i,j}$ th bone to the coordinate system $(XYZ)_{n_{i,j-1}}^1$ of the $n_{i,j-1}$ th bone, and L_i and $L_{n_{i,j}}$ are the lengths of the i th bone and the $n_{i,j}$ th parent bone, respectively. Similarly, we read the two sets $\{\mathbf{R}_{n_{i,j}, n_{i,j-1}}^1\}$ and $\{L_i\}$ ($L_{n_{i,j}} \in \{L_i\}$) from the database.

For $t > 1$, the position vector $\vec{r}_{P_2}^t$ of the post endpoint for the root bone, i.e., $i = 1$, is calculated by

$$\vec{r}_{P_2}^t = \vec{r}_{P_1}^t + T_1^t + \vec{d}_1^t \quad (8)$$

where T_1^t is the translation vector of the root bone at t with respect to $t = 1$, and $\vec{d}_1^t$ is calculated by

$$\vec{d}_1^t = \mathcal{T}_{1,\text{ref}}^t \mathbf{R}_1^t [0, L_1, 0]^T \quad (9)$$

where $\mathbf{R}_1^t$ is the recorded rotation matrix of the root bone at t with respect to $t = 1$.

Similarly, the position vectors $\vec{r}_{P_i}^t$ and $\vec{r}_{P_{i+1}}^t$ for $i > 1$ are calculated by

$$\begin{cases} \vec{r}_{P_i}^t = \vec{r}_{P_1}^t + \sum_{j=1}^{J_i} \vec{d}_{n_{i,j}}^t \\ \vec{r}_{P_{i+1}}^t = \vec{r}_{P_i}^t + \sum_{j=1}^{J_i} \vec{d}_{n_{i,j}}^t + \vec{d}_i^t. \end{cases} \quad (10)$$

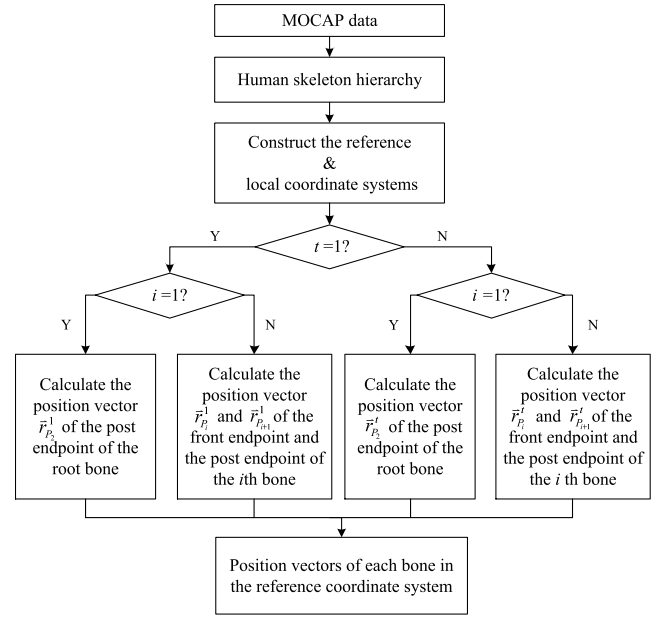


Fig. 4. Flowchart for the calculation of position vectors in the reference coordinate system.

In (10), $\vec{d}_i^t$ is calculated by

$$\begin{cases} \vec{d}_i^t = \mathcal{T}_{1,\text{ref}}^t \prod_{j=2}^{J_i} \mathbf{R}_{n_{i,j}, n_{i,j-1}}^t \mathbf{R}_i^t [0, L_i, 0]^T \\ \vec{d}_{n_{i,j}}^t = \mathcal{T}_{1,\text{ref}}^t \prod_{j'=2}^j \mathbf{R}_{n_{i,j'}, n_{i,j'-1}}^t \mathbf{R}_{n_{i,j}}^t [0, L_{n_{i,j}}, 0]^T \end{cases} \quad (11)$$

where $\mathbf{R}_{n_{i,j}, n_{i,j-1}}^t$ is the coordinate transform matrix from the local coordinate system of the $n_{i,j}$ th bone to the local coordinate system of the $n_{i,j-1}$ th bone at t , and $\mathbf{R}_i^t$ and $\mathbf{R}_{n_{i,j}}^t$ are the rotation matrices of the i th bone and the $n_{i,j}$ th parent bone at t with respect to $t = 1$, respectively.

Given $\vec{r}_{P_i}^t$ and $\vec{r}_{P_{i+1}}^t$, we could obtain the position vectors of the front and post endpoints as well as the direction vector $\vec{d}_i^t$ in the reference coordinate system according to (2)–(4) and (8), which lay foundations for the calculation of the instantaneous slant range of each bone during radar echo generation. The whole process is shown in Fig. 4.

C. Radar Echo Generation

To generate radar echoes of various activities, we set the radar observation coordinate system UVW to be parallel to the reference coordinate system $X_r Y_r Z_r$. As shown in Fig. 5, the pitch angle β_R is the angle between radar line of sight (LOS) l_{los} and its projection onto the $UO'W$ plane, while the azimuth angle α_R is the angle between the radar LOS and the $O'W$ -axis. Therefore, the unit vector of radar LOS is determined by

$$\hat{l}_{\text{los}} = [\cos \alpha_R \cos \beta_R, \sin \alpha_R \cos \beta_R, \sin \beta_R]. \quad (12)$$

Because strong m-D is produced when the human body is moving along l_{los} , we set l_{los} to be parallel to the motion direction. In addition, the pitch angle is set to 0° during observation.

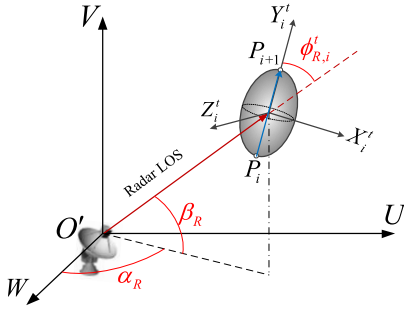
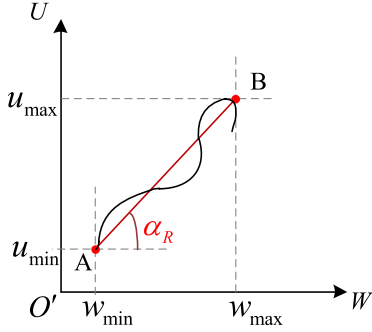


Fig. 5. Radar observation coordinate system.

Fig. 6. Trajectory of human motion on the $UO'W$ plane.

Specifically, the black curve between A and B in the $UO'W$ plane shown in Fig. 6 denotes the trajectory of human motion. Then, for the two sets $\{u_{\min}, w_{\min}\}$ and $\{u_{\max}, w_{\max}\}$, i.e., the maximum and minimum U and V coordinates for all the bone endpoints, the azimuth angle α_R is calculated by

$$\alpha_R = \tan^{-1} \frac{u_{\max} - u_{\min}}{w_{\max} - w_{\min}}. \quad (13)$$

The radar cross section (RCS) of the i th bone at t is approximated by the RCS of an ellipsoid, which is calculated by

$$\sigma_i^t = \frac{\pi a_i^2 b_i^2 c_i^2}{(a_i^2 \sin^2 \phi_{R,i}^t \cos^2 \alpha_R + b_i^2 \sin^2 \phi_{R,i}^t \sin^2 \alpha_R + c_i^2 \cos^2 \phi_{R,i}^t)^2} \quad (14)$$

where a_i , b_i , and c_i are the three semimajor axes of the i th bone, and $\phi_{R,i}^t$ is the incident angle at t , i.e., the angle between the Y'_i -axis and the radar LOS. Given $\vec{d}_i^t$, $\phi_{R,i}^t$ can be calculated by

$$\phi_{R,i}^t = \cos^{-1} \frac{\langle \vec{d}_i^t, \hat{l}_{\text{los}} \rangle}{\|\vec{d}_i^t\|_2} \quad (15)$$

where $\langle \cdot \rangle$ is the inner product and $\|\cdot\|_2$ is the L_2 -norm. The instantaneous slant range of the i th bone at t is the projection of $\vec{m}_i^t$ along the radar LOS, that is

$$R_i^t = \langle \vec{m}_i^t, \hat{l}_{\text{los}} \rangle \quad (16)$$

where $\vec{m}_i^t$ is calculated by (3).

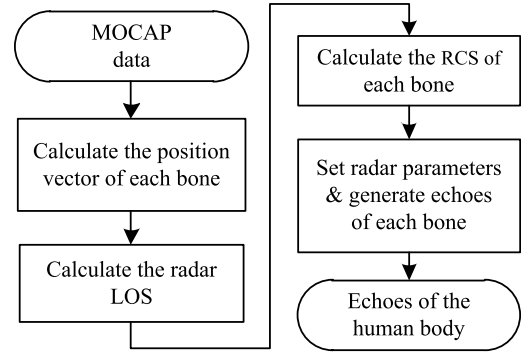


Fig. 7. Flowchart for radar echo generation based on the MOCAP data set.

For narrowband radar, echoes of the i th bone satisfy

$$s_i(t) = a_i^t e^{-j \frac{4\pi f_c}{c} R_i^t} \quad (17)$$

where f_c is the carrier frequency, C is the speed of light, and a_i^t is the time-varying amplitude with $a_i^t = \sqrt{\sigma_i^t}$. Then, echoes of the whole body are calculated by

$$s(t) = \sum_{i=1}^{N_b} a_i^t e^{-j \frac{4\pi f_c}{c} R_i^t} \quad (18)$$

where $N_b = 30$ is the number of bones and the m-D modulation is induced by the time-varying dR_i^t/dt .

The flowchart for radar echo generation from the MOCAP data set is shown in Fig. 7.

III. HUMAN GAIT CLASSIFICATION BASED ON DC-DCNN

As discussed in Section I-B, applying the STFT images with single fixed sliding window length as training and test samples is inadequate for accurate description of the motion characteristics for the torso and limbs. On the other hand, selecting multiple windows with equal intervals may generate similar STFT images, which increases the sample redundancy and generates a large number of network parameters.

In this section, we will study the choice of proper window length based on the nonstationary characteristics and motion periods of torso and limbs. Then, we will present detailed structures of the proposed DC-DCNN network and discuss the training process.

A. Choice of the STFT Window Length for Dual Channels

For the recognition of human activities with distinguishable periods, e.g., walking, running, jumping, climbing, and so on, the STFT samples with single fixed window length are adequate for obtaining satisfying recognition results [28], [30], [32]. In this paper, we will focus on refined human gait recognition, whose recognition accuracy is more sensitive to the choice of the window length.

The STFT of $s(t)$ is defined as

$$\text{STFT}(t, \omega) = \int s(t') w(t' - t) \exp\{-j\omega t'\} dt' \quad (19)$$

where $w(\cdot)$ is the window function, whose length determines the time and frequency resolution of the JTF distribution [38]; t and ω denote the time and frequency grids, respectively.

Generally, in one motion cycle, i.e., one motion period of the torso, the human gaits include the following phases: standing on feet, stepping out with one foot (swinging), double feet support, stepping out with the other foot (swinging), and finally, go back to the standing-on-feet state and enter the next motion cycle. Consequently, all types of gaits include two major stages in one motion cycle: the stage of swinging the left leg and the stage of swinging the right leg; the articulation action of the two stages is the double feet supporting action. Based on the above-mentioned analysis, the limbs swing twice in one motion cycle of the torso, and thus, the motion period of the torso T_t is about twice longer than the motion period of limbs T_l . Accordingly, we select two sliding windows with different lengths to generate samples for the dual channels: the first one is shorter to enhance features of the limbs, while the second one is longer to enhance features of the torso. Similar to human gait recognition, variational window length is also effective in capturing fine features of the body segments in other refined human activity recognition applications, as will be illustrated in Section IV-D.

Then, samples of the dual channels will be further processed and enhanced according to the statistical characteristics of JTF images and experimental results as follows.

- 1) For STFT images with shorter window length, calculate the mean of each image, i.e., $T_h = (I_{\min} + I_{\max})/2$, where $I_{\min}$ and $I_{\max}$ denote the minimum and maximum pixel values, respectively. Then, set the pixel values larger than or equal to T_h as T_h . By this means, we mitigate the influence of strong torso reflections on the feature extraction of limbs.
- 2) For STFT images with longer window length, take the power transform with the power of 1.2 to enhance the reflection of the torso.

In addition, we subsample the large STFT images, i.e., to resize the original STFT image by linear interpolation and resampling to reduce the number of network parameters and alleviate overfitting [39].

B. Proposed Network Architecture

As shown in Fig. 8, the proposed DC-DCNN network consists of dual channels, whose inputs are STFT images with longer and shorter sliding windows of human radar echoes, respectively. In addition, they share the same network topology, i.e., four convolutional layers, three pooling layers, and one fully connected layer. Then, hierarchical features of the dual channels are merged through the concatenation layer and the last fully connected layer. Finally, the classification result is given by the *softmax* classifier. Such a network structure guarantees that weights of the dual channels share the same dimension. In particular, the weights of each channel are updated independently to fully exploit features of the torso and limbs.

Considering the symmetric structure of DC-DCNN, we will only provide detailed settings of Channel 1 in the

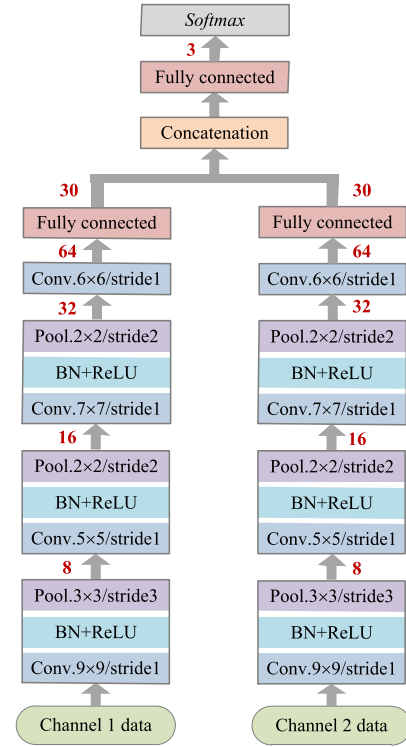


Fig. 8. Structure of DC-DCNN.

following discussions. The size of the input STFT image is 128×128 . To extract preliminary features, the kernel size of the first convolutional layer is set to 9×9 , generating eight feature maps with size 120×120 . Then, these feature maps are subsampled to 40×40 by the first pooling layer with kernel size 3×3 and stride size 3. The kernel size of the second convolutional layer is 5×5 , generating 16 36×36 feature maps, which are then subsampled to 18×18 by the second pooling layer with kernel size 2×2 and stride size 2. The third convolutional layer contains 32 filters with size 7×7 , yielding 32 feature maps with size 12×12 . Then, they are subsampled to 6×6 by the third pooling layer with kernel size 2×2 and stride size 2. The fourth convolutional layer consists of 64 filters with size 6×6 ; therefore, finally, the size of the 30 feature maps is 1×1 . In addition, a batch normalization (BN) layer is added after each convolutional layer.

In the following, we will provide detailed descriptions of the convolutional layer, the BN layer, the concatenation layer, and the *softmax* classifier of the proposed network. During forward propagation, the output of the convolutional layer can be expressed as

$$c_j^l = f \left(\sum_i (x_i^{l-1} * W_{ij}^l) + b_j^l \right) \quad (20)$$

where c_j^l is the j th feature map in the l th layer, $f(\cdot)$ is the nonlinear activation function rectified linear unit (ReLU) [40], x_i^{l-1} is the i th feature map in the $l-1$ th layer, $*$ is the convolution operation, W_{ij}^l is the weight between the i th image and the j th feature map, and b_j^l is the bias of the j th

feature map. If the size of the feature map is $n \times n$ in the $l-1$ th layer and the kernel size is $m \times m$ with stride size of 1, then the size of the output feature map in the l th layer is $(n-m+1) \times (n-m+1)$. To reduce feature redundancy and the number of network parameters, we apply max-pooling [41] after each convolutional layer.

In practice, it is highly possible that the training and test data sets have different distributions, which could decrease the convergence speed and degrade the generalization performance. In addition, the convolution and pooling operations may also change the data distribution. To tackle this problem, the proposed DC-DCNN structure adds a BN layer behind each convolutional layer for data normalization [42]. Supposing that a minibatch contains M feature maps with M activations, then for the set of feature maps $\mathbf{a}_B = \{a_i\}$, $i \in [1, M]$, the BN layer calculates their mean and variance, respectively

$$\mu_B = \frac{1}{M} \sum_{i=1}^M a_i \quad (21)$$

$$\sigma_B^2 = \frac{1}{M} \sum_{i=1}^M (a_i - \mu_B)^2 \quad (22)$$

and performs normalization by

$$\hat{a}_i = \frac{a_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}} \quad (23)$$

where ε is a small nonzero constant.

Then, it performs translation and scaling, that is

$$y_i = \gamma_0 \hat{a}_i + \beta_0. \quad (24)$$

In this paper, we set $\gamma_0 = 1$ and $\beta_0 = 0$, which remain constant during network training.

If we denote the output vector of the fully connected layer as $\mathbf{F}^1 = (f_1^1, f_2^1, \dots, f_{N_1}^1)$ and $\mathbf{F}^2 = (f_1^2, f_2^2, \dots, f_{N_2}^2)$, respectively, for Channels 1 and 2, then the concatenation layer performs the concatenation operation defined by

$$\mathbf{o} = \mathbf{F}^1 \oplus \mathbf{F}^2 \triangleq (f_1^1, f_2^1, \dots, f_{N_1}^1, f_1^2, f_2^2, \dots, f_{N_2}^2) \quad (25)$$

where $N_1 = N_2$ is the number of elements in the vector. Obviously, this operation stacks vectors of the two channels.

For multiclass classification, the *softmax* function is added behind the fully connected layer to compute the probability of the n th sample $\mathbf{x}_n$, $n \in [1, N]$, to be classified to the y_n th category [39], that is

$$p_n = \frac{e^{o_{y_n}}}{\sum_{k=1}^K e^{o_k}}, \quad \mathbf{o}_k = \mathbf{W}_k^T \mathbf{x}_n, \quad \mathbf{o}_{y_n} = \mathbf{W}_{y_n}^T \mathbf{x}_n \quad (26)$$

where $k, y_n \in [1, K]$, and K is the number of categories; $\mathbf{W}_k$ and $\mathbf{W}_{y_n}$ are the weights of the k th node and y_n th node in the last layer, respectively.

Then, the cross-entropy loss function is defined by

$$L = \frac{1}{N} \sum_n L_n = \frac{1}{N} \sum_n (-\log p_n). \quad (27)$$

By minimizing the loss function, the adjustable network parameters are gradually inclined to values that maximize

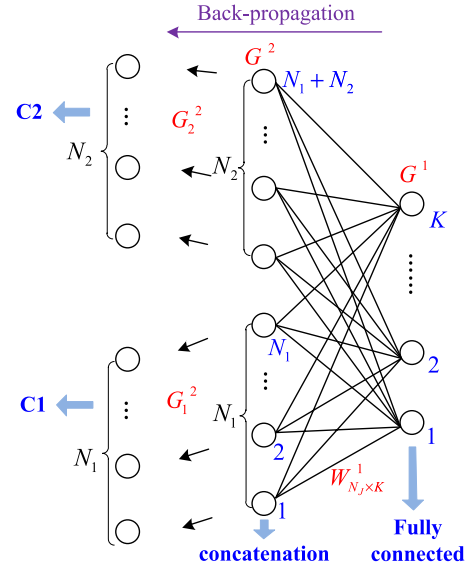


Fig. 9. Illustration of backpropagation for the concatenation layer.

the recognition accuracy during training. Then, in the testing process, the *softmax* layer calculates the probability of each category and then selects the category with the highest probability as the correct class label.

C. Network Training

According to Fig. 8, all the layers of the proposed network are directly connected and the error can be backpropagated to the entire network to update the weights. Therefore, the DC-DCNN is end-to-end trainable. In this paper, we train it according to the forward-backward propagation method [39] and update the network parameters by minimizing the loss function via gradient descent.

In particular, for the concatenation layer, supposing that the number of nodes in Channels 1 and 2 is N_1 and N_2 , respectively, as shown in Fig. 9. Then, the gradient of the loss function J with respect to activations for the concatenation layer during backpropagation is calculated to be

$$\mathbf{G}^{(2)} = \nabla_H J = \mathbf{G}^{(1)} (\mathbf{W}_{N_J \times K}^{(1)})^T \quad (28)$$

where $\mathbf{G}^{(1)}$ is the gradient of the fully connected layer, $\mathbf{W}_{N_J \times K}^{(1)}$ is the weight matrix between the concatenation layer and the fully connected layer, $N_J = N_1 + N_2$ is the number of nodes for the concatenation layer, and K is the number of nodes for the fully connected layer, i.e., the number of categories. Then, we divide $\mathbf{G}^{(2)}$ into $\mathbf{G}_1^{(2)}$ and $\mathbf{G}_2^{(2)}$, i.e., $\mathbf{G}^{(2)} = [\mathbf{G}_1^{(2)}, \mathbf{G}_2^{(2)}]$ and assign them to Channels 1 and 2, respectively, to complete the error backpropagation from the fully connected layer to the concatenation layer.

During training, we compute the derivative of the loss function L with respect to the trainable parameters layer by layer and then update them by the stochastic gradient descent with momentum and weight decay [43]. The momentum, which combines the gradient of the current iteration with the gradient of the previous iteration, is an effective optimization technique in gradient descent. Given the weight variation $\Delta \mathbf{w}^{(t)}$

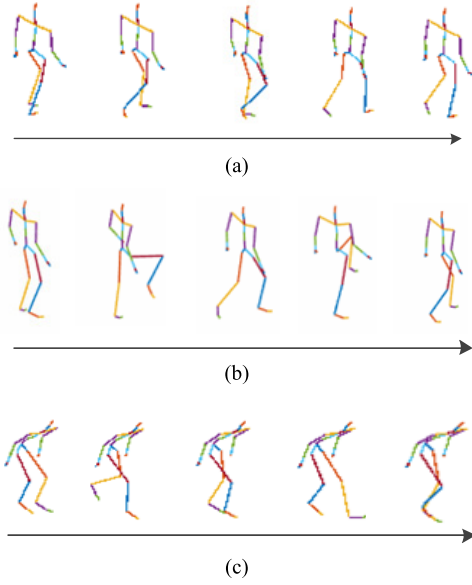


Fig. 10. Schematics of the three gaits. (a) Normal walking. (b) High-leg walking. (c) Bending walking.

at the τ th iteration, the new weight $w^{(\tau+1)}$ is updated by

$$\Delta w^{(\tau+1)} = \alpha \cdot \Delta w^{(\tau)} - \beta \eta_{\tau} w^{(\tau)} - \eta_{\tau} \left\langle \frac{\partial L}{\partial w} \bigg|_{w^{(\tau)}} \right\rangle \quad (29)$$

$$w^{(\tau+1)} = w^{(\tau)} + \Delta w^{(\tau+1)} \quad (30)$$

where $\alpha = 0.9$ is the momentum, $\beta = 0.9$ is the weight decay coefficient, and η_{τ} is the learning rate.

The updating formula of the learning rate satisfies $\eta_{\tau} = \eta_0 \cdot \gamma^{\text{floor}(\tau/\text{step})}$, where the initial learning rate η_0 is set to 0.02, the parameters γ and step are set to 0.1 and 100, respectively, and τ is the index of iteration. By this means, the learning rate decreases to 10% of its previous value per 100 iterations.

In addition, as computing the gradient of the loss function with respect to all the training samples has high computational complexity, we apply the minibatch gradient descent, which only utilizes a small part of the training examples at one time (e.g., 50 samples), to facilitate computation [41].

IV. EXPERIMENTAL RESULTS

A. Data Set Generation

For refined human gait recognition, we generate radar echoes of three typical gaits, i.e., normal walking, high-leg walking, and bending walking from the MOCAP database. Compared with normal walking, high-leg walking has different limb movements, while bending walking has different torso movements, as shown in Fig. 10, respectively. Accordingly, the corresponding STFT images of the three gaits have high similarity, which would degrade the performance of SC-DCNN.

For radar echo generation, the carrier frequency f_c is set to 10 GHz and the pulse repetition frequency is set to 2400 Hz. The azimuth angle is determined by the motion trajectory according to (13), and the pitch angle is set to 0° . For each gait,

TABLE I
SAMPLE SETTINGS FOR THE DC-DCNN

Types of gaits	Normal walking		High-leg walking		Bending walking	
	Channel index		Channel index		Channel index	
Channel index	1	2	1	2	1	2
Sliding window length	101	201	101	201	101	201
No. of training samples	200		150		160	
No. of test samples	100		50		100	
No. of samples	300		200		260	

the observation time of a sample series is 2 s. For DC-DCNN, the length of the sliding windows for Channels 1 and 2 is set to 101 and 201 according to the nonstationary characteristics of torso and limbs, respectively. The number of training and test samples is listed in Table I. For this data set, the sample imbalance causes little impact on network training and recognition performance evaluation because the three sample sets share the same magnitude.

To reduce the computational burden while maintaining the key JTF features, all the STFT images are subsampled to 128×128 . In addition, we preprocess all the samples following the method introduced in Section III-A and then subtract the mean of training samples to make the network stable and easy to converge [44]. Typical samples of the three gaits are shown in Fig. 11. For the STFT images with shorter sliding window length [Fig. 11(a)–(c)], it is observed that the JTF features of limbs are enhanced, while the JTF features of the torso become obvious by applying longer sliding window, as shown in Fig. 11(d)–(f). Because high-leg walking usually generates leg movements with large amplitude, the limbs shown in Fig. 11(b) have larger m-D bandwidth than normal walking shown in Fig. 11(a). In addition, bending walking and normal walking correspond to different torso m-D features, as shown in Fig. 11(d) and (f), respectively. Accordingly, the dual-channel processing with different sliding window lengths enriches the STFT samples and exploits more distinguishable features compared with single-channel processing, which could improve the performance on refined gait recognition.

B. Classification Results and Algorithm Comparison

Considering the limited number of training and test samples generated from the MOCAP data set, we do tenfold cross-validation by selecting them cyclically [39], and finally, the average recognition accuracy is 96.24% for the proposed DC-DCNN structure, as shown by the second column in Table II.

For comparison purpose, the optimal SC-DCNN with similar structural parameters is designed, as shown in Fig. 12. Then, the STFT samples with longer and shorter sliding windows serve as inputs of the SC-DCNN separately to compute the recognition accuracy, and the corresponding average accuracy is 92.84% and 89.60%, respectively, for the two types of samples, as shown by the third and fourth columns in Table II.

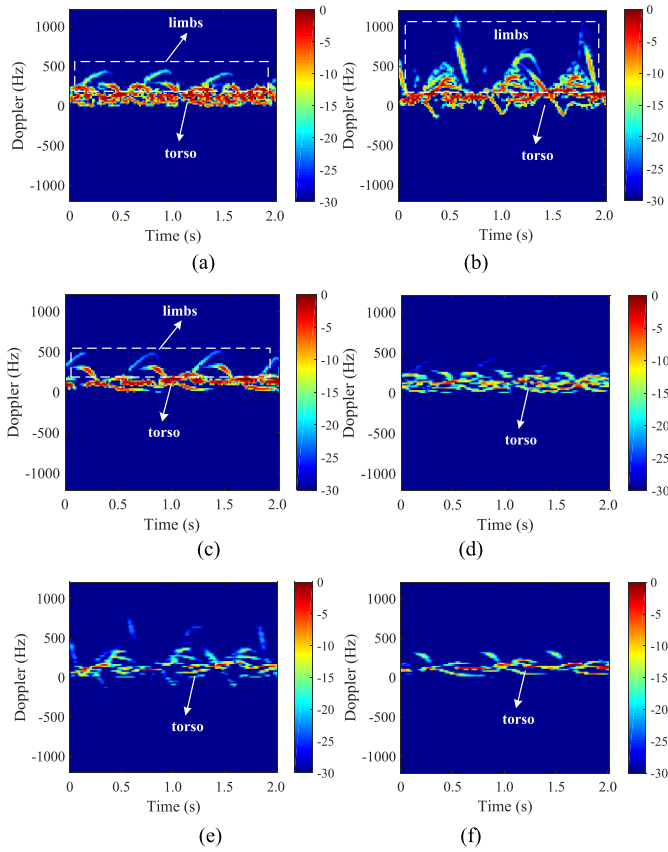


Fig. 11. Typical samples of three gaits. (a)–(c) Samples of normal walking, high-leg walking, and bending walking of Channel 1 (shorter sliding window). (d)–(f) Samples of normal walking, high-leg walking, and bending walking of Channel 2 (longer sliding window).

TABLE II
RECOGNITION RESULTS OF CROSS VALIDATION

Method Acc. (%)	DC-DCNN	SC-DCNN with shorter sliding window	SC-DCNN with longer sliding window	SC-DCNN with mixed samples
CV1	93.20	86.00	83.60	86.60
CV2	98.00	92.80	94.80	92.40
CV3	96.80	96.00	92.40	94.80
CV4	99.20	96.00	91.60	94.00
CV5	96.00	94.40	87.20	95.00
CV6	97.20	96.00	88.40	91.80
CV7	92.40	90.40	88.40	90.80
CV8	98.80	90.40	90.00	81.40
CV9	95.60	94.80	94.00	91.80
CV10	95.20	91.60	85.60	84.80
Average Acc.	96.24	92.84	89.60	90.34

To demonstrate that the proposed DC-DCNN achieves higher recognition accuracy by enhancing and fusion of features for both the torso and limbs in the dual channels, rather than by increasing the number of training samples, we design the third experiment for SC-DCNN. In this experiment, the samples with shorter and longer sliding windows are mixed up to form the training and test sets, and the average recognition accuracy is 90.34%.

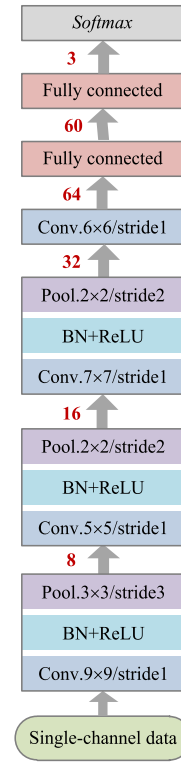


Fig. 12. Structure for SC-DCNN.

TABLE III
RECOGNITION RESULTS WITH VARYING SNR

Method Acc. (%)	DC-DCNN	SC-DCNN with shorter sliding window	SC-DCNN with longer sliding window	SC-DCNN with mixed samples
0dB	82.40	81.60	78.40	78.00
5dB	83.20	81.60	76.00	78.60
10dB	90.80	88.00	82.40	83.00

The recognition results demonstrate that the proposed DC-DCNN outperforms SC-DCNN and achieves higher recognition accuracy by extracting features independently of dual channels. In addition, for SC-DCNN, the recognition accuracy for mixed samples is between that for samples with shorter sliding window and samples with longer sliding window, indicating that the easily misclassified samples will decrease the recognition accuracy.

C. Robustness to Additive Noise

In real-world situations, the test samples are usually contaminated by additive Gaussian noise. To evaluate the robustness of DC-DCNN to noisy samples, we add Gaussian noise to the original noise-free test samples and carry out experiments with the same settings and parameters as those adopted in Section IV-B. The average recognition accuracy for the SNR of 0, 5, and 10 dB is given in Table III. It is observed that the proposed DC-DCNN achieves the highest recognition accuracy for each SNR.

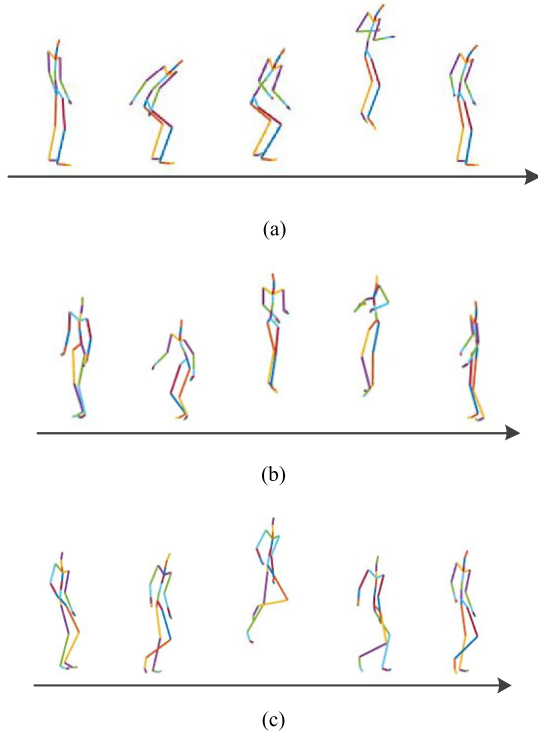


Fig. 13. Schematics of three types of jumping motion. (a) Normal jumping. (b) Turn-back jumping. (c) One-legged jumping.

D. Extended Experiment to Other Refined Human Activity Recognition

The above-mentioned experiments have demonstrated that DC-DCNN is effective for refined human gait recognition and could achieve higher recognition accuracy than SC-DCNN. To show that the proposed network could be readily extended to the recognition of other refined human activities, we generate radar echoes of three typical types of human jumping, i.e., normal jumping, turn-back jumping, and one-legged jumping from the MOCAP database according to the method introduced in Section II. Compared with normal jumping, the turn-back jumping has different torso movements, while one-legged jumping has different limb movements, as shown in Fig. 13(a)–(c), respectively.

For radar echo generation, the carrier frequency f_c is set to 9 GHz and the pulse repetition frequency is set to 3600 Hz to avoid Doppler aliasing. The azimuth angle is determined by the motion trajectory according to (13), and the pitch angle is set to 0° . For each type of jumping, the observation time of a sample series is 0.4 s. Because of the highly nonstationary jumping motion, we set the sliding window length for Channels 1 and 2 to 31 and 201, respectively. Detailed information of the training and test sets is listed in Table IV.

Similar to the previous experiments, all the STFT images are subsampled to 128×128 to reduce overfitting and the computational burden. For STFT images with shorter window length, we calculate the mean of each image, i.e., $T_h = (I_{\min} + I_{\max})/2$, where $I_{\min}$ and $I_{\max}$ denote the minimum and maximum values, respectively, and then set pixel values larger

TABLE IV
SAMPLE SETTINGS OF JUMPING FOR DC-DCNN

Types of jumping	Normal jumping		Turn-back jumping		One-legged jumping	
Channel index	1	2	1	2	1	2
Sliding window length	31	201	31	201	31	201
No. of training samples	100		100		115	
No. of test samples	85		80		90	
No. of samples	185		180		205	

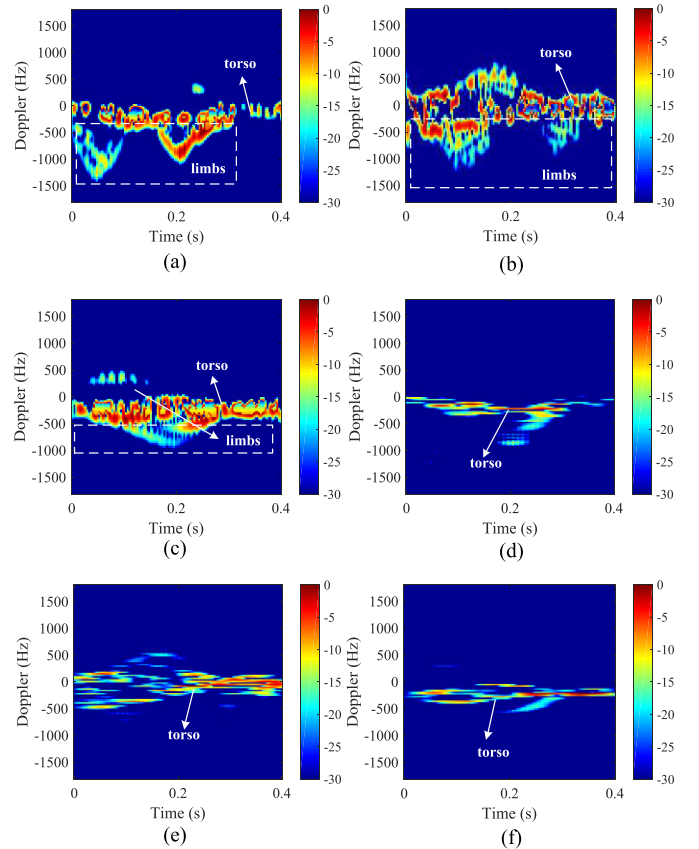


Fig. 14. Typical samples of three types of jumping. (a)–(c) Samples of normal jumping, turn-back jumping, and one-legged jumping of Channel 1 (shorter sliding window). (d)–(f) Samples of normal jumping, turn-back jumping, and one-legged jumping of Channel 2 (longer sliding window).

than or equal to T_h as $0.5T_h$. For STFT images with longer window length, we take the power transform with the power of 1.5 to enhance a JTF image of the torso. Then, we subtract the mean of training samples to make the network stable and easy to converge.

Typical samples of the three types of jumping are shown in Fig. 14, where Fig. 14(a)–(c) presents the STFT images with shorter sliding window length and Fig. 14(d)–(f) presents the STFT images with longer sliding window length. As one-legged jumping and normal jumping differ in leg movements, the JTF characteristics of limbs shown in Fig. 14(c) differ from those shown in Fig. 14(a). In addition, the torso movements of turn-back jumping exhibit wider Doppler bandwidth than

TABLE V
RECOGNITION RESULTS OF THE JUMPING MOTION

Method	DC-DCNN	SC-DCNN with shorter sliding window	SC-DCNN with longer sliding window	SC-DCNN with mixed samples
Acc. (%)	94.12	90.20	85.10	86.47

one-legged jumping and normal jumping, as illustrated by the torso characteristics shown in Fig. 14(d)–(f).

The recognition results given in Table V are obtained by the same network structure and parameter settings as those adopted in Sections IV-B and IV-C. It is observed that for the jumping motion, the recognition accuracy of the proposed DC-DCNN structure is about 8% higher than that of SC-DCNN, indicating that the proposed method can be readily extended to refined recognition of other types of human activities.

V. CONCLUSION

This paper proposed an effective radar echo generation method of human gaits from the infrared MOCAP data and designed the DC-DCNN for refined human gait recognition. In particular, the proposed network could extract enhanced JTF features of both torso and limbs separately and achieve effective feature fusion. Experimental results have shown that the recognition accuracy of the proposed DC-DCNN is higher than that of SC-DCNN. In addition, the proposed network has shown robustness to low SNR scenarios and is ready to be extended to other refined human activity recognition problems by selecting proper sliding window length and designing proper subchannel structures.

Future work will be focused on effective radar-based feature extraction and recognition of human gaits in complex environments and on the designing of robust human activity recognition methods based on sequence images.

ACKNOWLEDGMENT

The authors would like to thank all the anonymous reviewers and editors for their useful comments and suggestions that greatly improved this paper.

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